CEVE 421/521 Schedule

Note

Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.

Final Project: Deliverables are due throughout the semester (see schedule below).

CEVE 521 students: You will lead one Paper Discussion session. See Graduate Seminar Requirements.

Week	Date	What
1	Mon., Jan. 12	**Lecture:** Intro to Climate Risk Management slides
	Wed., Jan. 14	**Topical Discussion:** T. Frank [1] and S. Crawford [2]
	Fri., Jan. 16	**Lab 1:** Julia Setup & Hello World
2	Mon., Jan. 19	*Martin Luther King, Jr. Day (No Class)*
	Wed., Jan. 21	**Lecture:** Hazard, Exposure, Vulnerability slides *(in prep)*
	Fri., Jan. 23	**Lab 2:** Intro to iCOW *(in prep)*
3	Mon., Jan. 26	**Lecture:** Climate Variability, Extremes, and Change slides *(in prep)*
	Wed., Jan. 28	**Paper Discussion:** Climate Extremes (IPCC) *(in prep)*
	Fri., Jan. 30	**Lab 3:** Sea-Level Rise Scenarios with BRICK *(in prep)* / **Project:** Topic Proposal Due
4	Mon., Feb. 2	**Lecture:** Uncertainty & Monte Carlo slides *(in prep)*
	Wed., Feb. 4	**Paper Discussion:** Monte Carlo & Fat Tails *(in prep)*
	Fri., Feb. 6	**Lab 4:** Monte Carlo Hazard Modeling *(in prep)*
5	Mon., Feb. 9	**Lecture:** Risk Metrics (EAD, VaR, CVaR) slides *(in prep)*
	Wed., Feb. 11	**Lab 5:** Probabilistic Risk Analysis *(in prep)*
	Fri., Feb. 13	*No Class*

Week	Date	What
6	Mon., Feb. 16	**Exam 1** (Covers Module 1)
	Wed., Feb. 18	**Lecture:** Valuation & Benefit-Cost Analysis slides *(in prep)*
	Fri., Feb. 20	**Paper Discussion:** Discounting & BCA *(in prep)*
7	Mon., Feb. 23	**Lecture:** Optimal Static Decisions slides *(in prep)*
	Wed., Feb. 25	**Paper Discussion:** Optimal Decisions *(in prep)*
	Fri., Feb. 27	**Lab 6:** Simulation-Optimization *(in prep)* / **Project:** Memo 1 Due
8	Mon., Mar. 2	**Lecture:** Sensitivity & Value of Information slides *(in prep)*
	Wed., Mar. 4	**Paper Discussion:** Value of Information *(in prep)*
	Fri., Mar. 6	**Lab 7:** Value of Partial Information *(in prep)*
9	Mon., Mar. 9	**Lecture:** Trade-offs & Values slides *(in prep)*
	Wed., Mar. 11	**Paper Discussion:** Trade-offs and Values *(in prep)*
	Fri., Mar. 13	**Exam 2** (Covers Module 2)
	Mar. 16-20	*Spring Break*
10	Mon., Mar. 23	**Lecture:** Robustness & Deep Uncertainty slides *(in prep)*
	Wed., Mar. 25	**Paper Discussion:** Robustness & Deep Uncertainty *(in prep)*
	Fri., Mar. 27	**Lab 8:** Robustness Metrics *(in prep)* / **Project:** Memo 2 Due
11	Mon., Mar. 30	**Lecture:** Exploratory Modeling & Scenario Discovery slides *(in prep)*
	Wed., Apr. 1	**Paper Discussion:** Exploratory Modeling *(in prep)*
	Fri., Apr. 3	**Lab 9:** Scenario Discovery *(in prep)*
12	Mon., Apr. 6	**Lecture:** Sequential Decision Making slides *(in prep)*
	Wed., Apr. 8	**Paper Discussion:** Sequential Decision Making *(in prep)*

Week	Date	What
	Fri., Apr. 10	**Lab 10:** Real Options & Dynamic Policy Search *(in prep)* / **Project:** Memo 3 Due
13	Mon., Apr. 13	**Lecture:** Multi-Objective Optimization slides *(in prep)*
	Wed., Apr. 15	**Paper Discussion:** Multi-objective Optimization *(in prep)*
	Fri., Apr. 17	**Lab 11:** Pareto Fronts *(in prep)* / **Project:** Slides Due
14	Mon., Apr. 20	**Final Project Presentations** (Teams 1 & 2)
	Wed., Apr. 22	**Final Project Presentations** (Teams 3 & 4)
	Fri., Apr. 24	**Final Project Presentations** (Teams 5 & 6)

Bibliography

[1] T. Frank, "Bold New Jersey Shore Flood Rules Could Be Blueprint for Entire U.S. Coast," *Scientific American*, Aug. 2022, Accessed: Jan. 11, 2023. [Online]. Available: https://www.scientificamerican.com/article/bold-new-jersey-shore-flood-rules-could-be-blueprint-for-entire-u-s-coast/

[2] S. Crawford, "Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?." Accessed: Dec. 05, 2025. [Online]. Available: https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/